Cognitive Workload Allocation Under Uncertainty

A Stochastic Multiple Knapsack Approach with Workload Balance

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Agenda

Outline

Motivation: Cognitive Overload, Burnout

- Cognitive overload from excessive demands leads to billions of lost working days annually [Robinson, 2024].
- Productivity depends on **attention management** rather than rigid time scheduling [Grant, 2019].
- Traditional time management fails because it ignores the finite nature of human lifespan and cognitive capacity [Hart, 2024].
- Chronic illness, burnout risk, and work-family conflict underscore the need for systematic resource allocation that accounts for well-being [Cook and Zill, 2024, Yoo and Lee, 2022].

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- A workload-balance penalty $\lambda \geq 0$ discourages uneven utilization across agents.

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Auxiliary variables

- $y_j^{\omega} \in \{0, 1\}$: violation indicator
- $U_j \geq 0$: expected utilization
- $Z \geq 0$: max utilization gap

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The total violation probability for each agent must stay below α:

$$\sum_{\omega \in \Omega} p_\omega \, y_j^{\omega} \;\leq\; \alpha, \quad \forall j \in \mathcal{J}.$$

Workload Utilization and Balance

Expected workload utilization of agent j, normalized by capacity:

$$U_j \;=\; \frac{1}{C_j} \sum_{i \in \mathcal{I}} \sum_{\omega \in \Omega} p_\omega \, S_{ij}^{\omega} \, x_{ij}, \quad \forall j \in \mathcal{J}.$$

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- A smaller Z means a more balanced allocation.
- Normalization by C_j allows fair comparison across agents with different capacities.

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$$\max_{x,y,U,Z} \quad \underbrace{\sum_j \sum_i R_{ij}\, x_{ij}}_{\text{total return}} \qquad \text{(Obj)}$$

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The objective balances three forces: return creation, overload-risk control, and workload fairness.

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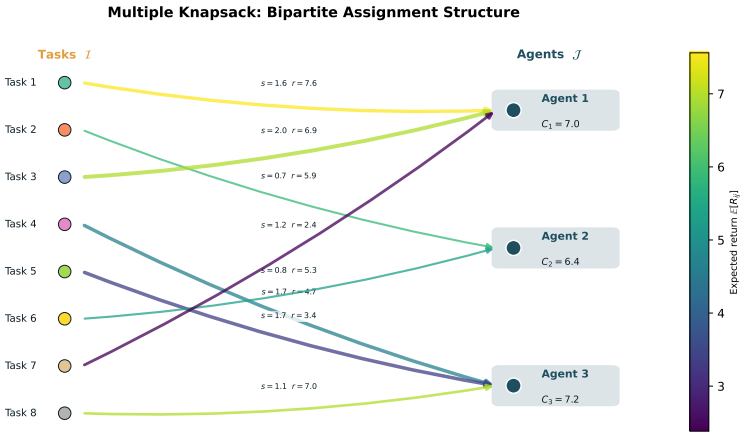
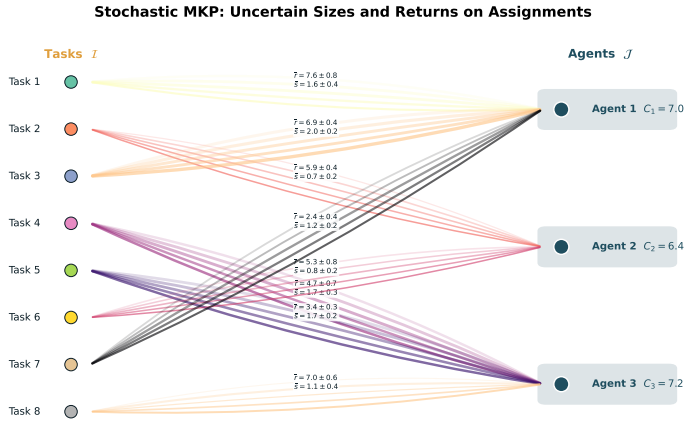
Multiple Knapsack Structure



Bipartite assignment: tasks (left) are assigned to agents (right) with capacities C_j. Edge width encodes task size $\mathbb{E}[S_{ij}]$ and color indicates expected return R_{ij}. Each task is assigned to at most one agent (constraint 10).

Stochastic Sizes and Returns



Stochastic extensions: each assignment edge shows multiple scenario realizations of size S_{ij}^{ω} (line width) and return R_{ij} (annotation). The chance constraint (10) limits the probability that total realized load exceeds agent capacity.

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Stochastic Knapsack: Cutting Planes and Dominance

- Büyüktahtakın [2023] derive **scenario dominance cuts** for multi-stage stochastic MIPs applied to the dynamic multi-dimensional knapsack. Strong dominance cuts reduce solution time with 0.06% average deviation from optimality.
- Büyüktahtakın [2022] generalize this via **Stage-t Scenario Dominance** for risk-averse (mean-CVaR) knapsack problems, achieving one-to-two orders of magnitude speedup.
- Joung et al. [2023] compare LP relaxation bounds for the robust knapsack problem and show that extended formulations yield identical bounds.

Distributionally Robust and Risk-averse Knapsack

- Bos et al. [2024] approximate the stochastic knapsack with periodic scheduled arrivals using **Wasserstein Distributed Robust Optimization (DRO)**, producing robust healthcare surgical schedules.
- Ding et al. [2022] propose a target-based DRO knapsack with **uncertain profit and capacity**, reformulated as a second-order conic program.
- Merzifonluoglu and Geunes [2021] study the **risk-averse static stochastic knapsack** with CVaR and mean-standard deviation objectives, deriving efficient heuristics for normally distributed resource consumption.
- Boonstra et al. [2025] solve a robust multi-item newsvendor with budget constraint using a greedy knapsack algorithm under moment ambiguity.

Solution Algorithms for Stochastic Programs

- Chen and Luedtke [2022] analyze **sample average approximation** for two-stage programs without complete recourse, proving exponential convergence of recourse likelihood.
- Larsen et al. [2024] accelerate L-shaped methods via **supervised ML**, solving stochastic multi-knapsack instances in 20% of the time with less than 0.1% optimality gap.
- Lefebvre et al. [2024] reformulate adjustable robust optimization with **discrete uncertainty** and apply branch-and-cut to the multiple knapsack problem with uncertain weights.
- Lyu et al. [2022] unify responsive, adaptive, and anticipative allocation policies for the stochastic knapsack through **persistency values**.

Assignment, Scheduling, and Workforce Planning

- Rezaeian et al. [2024] model the **stochastic assignment of projects** to managers via multi-stage stochastic integer programming with heuristic algorithms for large instances.
- Nasirian et al. [2025] solve multiskilled workforce staffing and scheduling via **logic-based Benders decomposition**, achieving 29x speedup over direct MILP solving.
- Cavagnini et al. [2020] use two-stage stochastic programming for production planning under **uncertain learning rates**, with insights on cross-training and worker assignment.
- Novak et al. [2022] connect stochastic parallel machine scheduling to an **inflatable stochastic knapsack** pricing subproblem within branch-and-price.

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  - A **master problem** that decides assignments and evaluates the return-balance trade-off.
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- This mirrors the managerial logic: propose an assignment first, then verify safety under uncertainty.

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Cuts are generated iteratively rather than enumerated. Convergence is finite because the dual polyhedron has finitely many extreme points and rays.

Our Decomposition Structure

Notation. Define the expected workload coefficient:

$$\bar{S}_{ij} = \sum_{\omega \in \Omega} p_\omega \, S_{ij}^{\omega}, \quad \forall i \in \mathcal{I}, \; \forall j \in \mathcal{J}.$$

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- The **workload-balance component** depends only on expected workload, so it stays in the master problem.
- The **scenario-based chance feasibility** is naturally evaluated only after a candidate assignment is fixed.

Master Problem (Iteration t)

Let $\mathcal{B}^t$ denote the set of feasibility cuts generated so far.

$$\max_{x,U,Z} \quad \sum_j \sum_i R_{ij}\, x_{ij} \;-\; \lambda\, Z \quad \text{(MP-Obj)}$$

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The master keeps the true objective and balance structure. Chance constraints are *not* included here; they are checked in the feasibility subproblem.

Feasibility Evaluation Subproblem

Given a candidate assignment $\bar{x}$ from the master:

- **Step 1.** Compute realized workload in each scenario:

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- **Step 4.** Check: $\bar{x}$ is chance-feasible if and only if $P_j^{\text{viol}}(\bar{x}) \leq \alpha$ for all $j \in \mathcal{J}$.

This is a **logical feasibility check**, not another optimization problem. Once $\bar{x}$ is fixed, all quantities are computed directly.

Feasibility Cut Generation: No-Good Cut

Suppose agent j violates the chance constraint under candidate $\bar{x}$.

- Let $\mathcal{I}_j(\bar{x}) = \{i \in \mathcal{I} : \bar{x}_{ij} = 1\}$ be the **full bundle** of tasks assigned to agent j.

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The no-good cut ensures finite convergence but may require many iterations because it eliminates only one assignment at a time.

Feasibility Cut Generation: Stronger Cut

The stronger cut targets the **root cause** of the violation.

- Within the full bundle $\mathcal{I}_j(\bar{x})$, identify a **minimal offending subset** $\mathcal{B}_j(\bar{x}) \subseteq \mathcal{I}_j(\bar{x})$ such that

$$\sum_{\omega \in \Omega:\ \sum_{i \in \mathcal{B}_j(\bar{x})} S_{ij}^{\omega} > C_j} p_\omega \ > \ \alpha.$$

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- This subset alone already causes the chance violation. Add the cut:

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- Because $\mathcal{B}_j \subseteq \mathcal{I}_j$ and $|\mathcal{B}_j| \leq |\mathcal{I}_j|$, the stronger cut **dominates** the no-good cut: it eliminates *every* assignment that contains the offending subset, not just the one specific bundle.

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Practical effect

Fewer iterations are needed because each stronger cut removes a larger region of infeasible assignments from the master problem.

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Property	No-good	Stronger
Assignments eliminated	1	many
Subset size	$\lvert\mathcal{I}_j\rvert$ (full)	$\lvert\mathcal{B}_j\rvert \leq \lvert\mathcal{I}_j\rvert$
Convergence speed	slower	faster
Validity guarantee	yes	yes

Algorithm: Step-by-Step

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Convergence

Both cut types guarantee finite termination: each cut eliminates at least one infeasible assignment pattern, and the total number of binary patterns is finite. Stronger cuts accelerate convergence by eliminating multiple patterns per iteration.

Why This Decomposition Works

- **Exact objective in master**: every master solution is evaluated with respect to the exact return-balance trade-off.

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- **Simple subproblem**: the feasibility check is purely arithmetic (no LP or MIP to solve).
- **Managerial interpretation**: the planner proposes an assignment, then checks whether it is sufficiently safe under workload uncertainty.

Outline

Test Instances

Three instances with increasing size:

Instance	Tasks	Agents	Scenarios	α	λ
Small	5	3	3	0.20	4.0
Medium	10	5	5	0.20	5.0
Large	25	10	10	0.10	6.0

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- Capacities vary slightly across agents (e.g., $[10, 10, 11]$ for the small instance).
- The large instance uses a tighter risk tolerance ($\alpha = 0.10$).

Computational Summary

Instance	Objective	Return	Z	Iterations	Cuts	Gap
Small	51.50	53	0.376	2	1	0.00
Medium	77.11	79	0.377	52	220	0.00
Large	254.34	257	0.444	443	1958	0.00

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- All instances close the Benders gap to zero.
- The number of iterations and cuts increases substantially with instance size: from 2 iterations / 1 cut (small) to 443 iterations / 1958 cuts (large).

Small Instance: Assignment and Utilization

Assignment

Task	Agent
1	Agent 3
2	Agent 2
3	Agent 3
4	Agent 2
5	Agent 1

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Expected utilization

Agent	U_j
1	0.570
2	0.940
3	0.946

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- Solved in 2 iterations with 1 feasibility cut.
- The first master solution was infeasible because agent 1 received an unsafe bundle. One cut corrected this.

Medium Instance: Assignment and Utilization

Assignment

Task	Agent
2	Agent 2
3	Agent 3
4	Agent 4
5	Agent 5
7	Agent 1
10	Agent 4

Tasks 1, 6, 8, 9: **not assigned**.

Medium Instance: Assignment and Utilization

Assignment

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Tasks 1, 6, 8, 9: **not assigned**.

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Agent	U_j
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Tasks 1, 6, 8, 9: **not assigned**.

- 52 iterations and 220 feasibility cuts.
- Leaving some tasks unassigned is optimal: assigning them would violate chance constraints or create too much imbalance.

Large Instance: Assignment and Utilization

Assignment highlights

- 19 of 25 tasks are assigned.
- Tasks 5, 11, 12, 17, 23, 24: not assigned.
- Most agents receive 2 tasks; agent 7 receives only 1.

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Agent	U_j
1	0.786
2	0.722
3	0.730
4	0.722
5	0.739
6	0.747
7	0.355
8	0.748
9	0.785
10	0.755

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443 iterations and 1958 cuts. Feasible but not perfectly balanced. Agent 7 is the lightest outlier.

Observations from Results

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- **Zero Benders gap**: all instances are solved to proven optimality, confirming correctness of the decomposition.
- **Computational cost**: the explicit outer-loop Benders scheme becomes expensive for large instances with tight chance constraints, suggesting room for acceleration strategies.

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Key Takeaways

- The formulation models cognitive resource allocation as a **stochastic multiple knapsack** with scenario-based chance constraints and a workload-balance penalty.

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- **Benders decomposition** separates assignment decisions from chance-feasibility checks, keeping the exact objective in the master and using a simple logical subproblem.
- Computational results on three instances demonstrate correctness (zero gap) and reveal the scalability challenge as instance size grows.
- Future directions include stronger cut generation, scenario dominance [Büyüktahtakın, 2023], and ML-accelerated decomposition [Larsen et al., 2024].

Outline

References I

Guus Boonstra, Wouter J.E.C. van Eekelen, and Johan S.H. van Leeuwaarden. Robust Knapsack Ordering for A Partially-informed Newsvendor with Budget Constraint. *Operations Research Letters*, 58, 2025. doi: 10.1016/j.orl.2024.107202.

Hayo Bos, Richard J. Boucherie, Erwin W. Hans, and Gréanne Leeftink. Distributionally Robust Scheduling of Stochastic Knapsack Arrivals. *Computers and Operations Research*, 167, 2024. doi: 10.1016/j.cor.2024.106641.

Esra Büyüktahtakın. Scenario-dominance to Multi-stage Stochastic Lot-sizing and Knapsack Problems. *Computers and Operations Research*, 153, 2023. doi: 10.1016/j.cor.2023.106149.

I. Esra Büyüktahtakın. Stage-t Scenario Dominance for Risk-averse Multi-stage Stochastic Mixed-integer Programs. *Annals of Operations Research*, 309, 2022. doi: 10.1007/s10479-021-04388-3.

Rossana Cavagnini, Mike Hewitt, and Francesca Maggioni. Workforce Production Planning Under Uncertain Learning Rates. *International Journal of Production Economics*, 225, 2020. doi: 10.1016/j.ijpe.2019.107590.

References II

Rui Chen and James Luedtke. On Sample Average Approximation for Two-stage Stochastic Programs Without Relatively Complete Recourse. *Mathematical Programming*, 196: 719–754, 2022. doi: 10.1007/s10107-021-01753-9.

Alexandra Cook and Alexander Zill. Individual Health Status As A Resource: Analyzing Associations Between Perceived Illness Symptom Severity, Burnout, and Work Engagement Among Employees with Autoimmune Diseases. *Applied Psychology*, 73:990–1025, 2024. doi: 10.1111/apps.12464.

Jianpeng Ding, Liuxin Chen, Ginger Y. Ke, Yuanbo Li, and Lianmin Zhang. Balancing The Profit and Capacity Under Uncertainties: A Target-based Distributionally Robust Knapsack Problem. *International Transactions in Operational Research*, 29:760–782, 2022. doi: 10.1111/itor.13031.

References III

Adam Grant. Productivity isn't about time management. it's about attention management., March 2019. URL https://www.nytimes.com/2019/03/28/smarter-living/productivity-isnt-about-time-management-its-about-attention-management.html. Accessed: January 25, 2026.

Hanna Hart. Productivity and the hard truth about time management, January 2024. URL https://www.forbes.com/sites/hannahart/2024/01/25/productivity-and-the-hard-truth-about-time-management/. Accessed: January 25, 2026.

Seulgi Joung, Seyoung Oh, and Kyungsik Lee. Comparative Analysis of Linear Programming Relaxations for The Robust Knapsack Problem. *Annals of Operations Research*, 323:65–78, 2023. doi: 10.1007/s10479-022-05161-w.

Eric Larsen, Emma Frejinger, Bernard Gendron, and Andrea Lodi. Fast Continuous and Integer L-shaped Heuristics Through Supervised Learning. *Informs Journal on Computing*, 36: 203–223, 2024. doi: 10.1287/ijoc.2022.0175.

References IV

Henri Lefebvre, Enrico Malaguti, and Michele Monaci. Adjustable Robust Optimization with Discrete Uncertainty. *Informs Journal on Computing*, 36:78–96, 2024. doi: 10.1287/ijoc.2022.0086.

Guodong Lyu, Mabel C. Chou, Chung Piaw Teo, Zhichao Zheng, and Yuanguang Zhong. Stochastic Knapsack Revisited: The Service Level Perspective. *Operations Research*, 70: 729–747, 2022. doi: 10.1287/opre.2021.2173.

Yasemin Merzifonluoglu and Joseph Geunes. The Risk-averse Static Stochastic Knapsack Problem. *Informs Journal on Computing*, 33:931–948, 2021. doi: 10.1287/ijoc.2020.0972.

Araz Nasirian, Lele Zhang, Alysson M. Costa, and Babak Abbasi. Multiskilled Workforce Staffing and Scheduling: A Logic-based Benders' Decomposition Approach. *European Journal of Operational Research*, 323:20–33, 2025. doi: 10.1016/j.ejor.2024.11.033.

References V

Antonin Novak, Premysl Sucha, Matej Novotny, Richard Stec, and Zdenek Hanzalek. Scheduling Jobs with Normally Distributed Processing Times On Parallel Machines. *European Journal of Operational Research*, 297:422–441, 2022. doi: 10.1016/j.ejor.2021.05.011.

Aidin Rezaeian, Hamidreza Koosha, Mohammad Ranjbar, and Saeed Poormoaied. The Assignment of Project Managers to Projects in An Uncertain Dynamic Environment. *Annals of Operations Research*, 341:1107–1134, 2024. doi: 10.1007/s10479-024-05958-x.

Bryan Robinson. 4 steps to mitigate cognitive overload and career stress before 2025, November 2024. URL `https://www.forbes.com/sites/bryanrobinson/2024/11/02/4-steps-to-mitigate-cognitive-overload-and-career-stress-before-2025/`. Accessed: January 25, 2026.

Hye Jin Yoo and Hyeongsuk Lee. Critical Role of Information and Communication Technology in Nursing During The COVID-19 Pandemic: A Qualitative Study. *Journal of Nursing Management*, 30:3677–3685, 2022. doi: 10.1111/jonm.13880.